Course 6014A | Semester VI | Theory | 4 Credits

Renewable Energy and Environment

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6014A as Renewable Energy and Environment in Semester VI for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Hyderabad’s Sustainable Energy Technology course and NPTEL IIT Roorkee’s Energy Resources, Economics, and Sustainability course. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Energy audits, power plant designs, environmental impact assessments and energy policy decisions must be based on approved engineering designs, certified equipment specifications, current environmental regulations and competent professional review.

Verified source note: Verified sources support energy scenarios, fossil fuel impacts, hydropower, wind energy, solar thermal/PV, bioenergy, geothermal power, energy storage, economics and LCA. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Renewable Energy and Environment studies the transition from fossil fuels to sustainable energy sources and their environmental impacts. It develops the ability to analyze solar, wind, hydro, biomass and geothermal energy systems, evaluate their environmental benefits and challenges, understand energy economics and sustainability, and apply energy conservation principles while respecting the technical and regulatory boundaries of energy production.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□.

Prerequisite foundation

Environmental studies, basic physics, elementary chemistry, engineering mathematics and environmental awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the global energy scenario, the impact of fossil fuels and the need for renewable energy sources.
2. Explain solar energy systems, including photovoltaic and solar thermal technologies and their applications.
3. Explain wind, hydro and geothermal energy systems, including site selection and working principles.
4. Explain bioenergy, energy storage technologies, energy economics and the principles of life cycle assessment (LCA).

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain energy trends, fossil fuel impacts and the fundamentals of solar energy systems.	Understand
CO2	Explain the principles and applications of wind, hydro and geothermal energy technologies.	Understand
CO3	Explain bioenergy production, energy storage systems and the hydrogen energy economy.	Understand
CO4	Explain energy economics, sustainability principles and life cycle assessment of energy systems.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7-PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Energy Scenario and Solar Technologies	Global and Indian energy scenarios; impact of fossil fuels and climate change; introduction to renewable energy; solar radiation; solar thermal systems; solar photovoltaic (PV) technology and efficiency.	Approx. 14 hours	CO1	Module 1
2	Wind, Hydro and Geothermal Energy	Wind energy fundamentals; wind turbine design and performance; hydropower principles; hydroturbine selection; pumped hydro storage; geothermal energy sources and power systems.	Approx. 16 hours	CO2	Module 2
3	Bioenergy, Storage and Hydrogen Economy	Bioenergy and biofuels (feedstocks, production technologies); biogas and biodiesel; energy storage technologies (mechanical, electrochemical, thermal); fuel cells and the hydrogen energy economy.	Approx. 16 hours	CO3	Module 3
4	Energy Economics, Sustainability and LCA	Energy demand characteristics; energy storage parameters; energy economics and policy; sustainability principles in energy; Life Cycle Assessment (LCA) of energy systems; carbon capture and storage (CCS).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Energy Scenario and Solar Technologies

Global and Indian energy scenarios; impact of fossil fuels and climate change; introduction to renewable energy; solar radiation; solar thermal systems; solar photovoltaic (PV) technology and efficiency.

CO1 · Approx. 14 hours

Wind, Hydro and Geothermal Energy

Wind energy fundamentals; wind turbine design and performance; hydropower principles; hydroturbine selection; pumped hydro storage; geothermal energy sources and power systems.

CO2 · Approx. 16 hours

Bioenergy, Storage and Hydrogen Economy

Bioenergy and biofuels (feedstocks, production technologies); biogas and biodiesel; energy storage technologies (mechanical, electrochemical, thermal); fuel cells and the hydrogen energy economy.

CO3 · Approx. 16 hours

Energy Economics, Sustainability and LCA

Energy demand characteristics; energy storage parameters; energy economics and policy; sustainability principles in energy; Life Cycle Assessment (LCA) of energy systems; carbon capture and storage (CCS).

CO4 · Approx. 14 hours

MODULE 1

Energy Scenario and Solar Technologies

Global and Indian energy scenarios; impact of fossil fuels and climate change; introduction to renewable energy; solar radiation; solar thermal systems; solar photovoltaic (PV) technology and efficiency.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

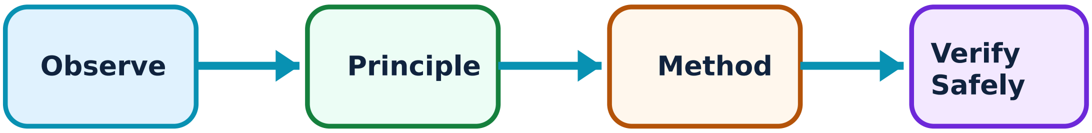
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Scenario and Solar Technologies: identify the system, state its principle, follow the correct method and verify the result safely.

1. Energy trends and fossil fuel impacts−

The world is shifting from carbon-intensive fossil fuels to renewable sources to mitigate climate change. Understanding macro trends in energy use (World and India) and the environmental consequences of fossil fuels is fundamental to the energy transition.

Study and application method

List three environmental impacts of fossil fuel use and compare the current renewable energy contribution in India vs the global average.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List three environmental impacts of fossil fuel use and compare the current renewable energy contribution in India vs the global average.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Energy data and climate projections are subject to frequent updates. Use current, authoritative reports (e.g., IEA, MNRE) for reliable energy planning.

2. Solar thermal and PV systems−

Solar energy is harnessed through thermal systems (water heaters, CSP) and Photovoltaic (PV) cells. PV technology converts sunlight directly into electricity using semiconductors. Efficiency depends on cell material, orientation and environmental factors.

Study and application method

Sketch a simple solar PV system and label the components; explain the difference between a solar thermal collector and a PV module.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Method: Sketch a simple solar PV system and label the components; explain the difference between a solar thermal collector and a PV module.

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Common mistake / precaution: Solar system performance is highly site-dependent. Do not design systems without authorised solar-path analysis and local weather data.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Wind, Hydro and Geothermal Energy

Wind energy fundamentals; wind turbine design and performance; hydropower principles; hydroturbine selection; pumped hydro storage; geothermal energy sources and power systems.

Approx. 16 hours **CO2** **Understand · Apply**

Learning goals

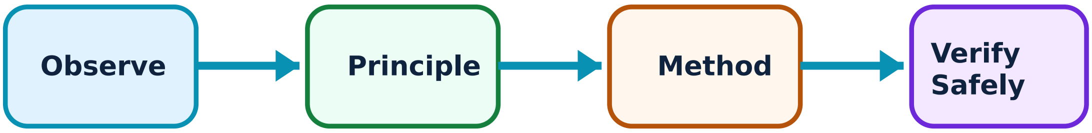
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wind, Hydro and Geothermal Energy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Wind and hydro power−

Wind energy uses turbines to convert kinetic energy into electricity, requiring sites with consistent wind speeds. Hydropower uses water flow to drive turbines, with pumped storage acting as a large-scale energy battery for grid stability.

Study and application method

Compare the working principles of a horizontal-axis wind turbine and a Pelton wheel hydroturbine; list the factors for selecting a wind-farm site.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the working principles of a horizontal-axis wind turbine and a Pelton wheel hydroturbine; list the factors for selecting a wind-farm site.

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Common mistake / precaution: Wind and hydro projects have significant land-use and ecological impacts. All projects must undergo authorised environmental and social impact assessments.

2. Geothermal energy systems−

Geothermal energy harnesses heat from the Earth's interior for power generation or direct heating. Systems include dry steam, flash steam and binary cycle plants, depending on the temperature and state of the geothermal resource.

Study and application method

Describe the three main types of geothermal power plants and identify the regions in India with potential for geothermal energy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the three main types of geothermal power plants and identify the regions in India with potential for geothermal energy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Geothermal extraction can trigger seismic activity or release underground gases. Exploration and production must follow strict safety and environmental protocols.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Bioenergy, Storage and Hydrogen Economy

Bioenergy and biofuels (feedstocks, production technologies); biogas and biodiesel; energy storage technologies (mechanical, electrochemical, thermal); fuel cells and the hydrogen energy economy.

Approx. 16 hours **CO3** **Understand · Apply**

Learning goals

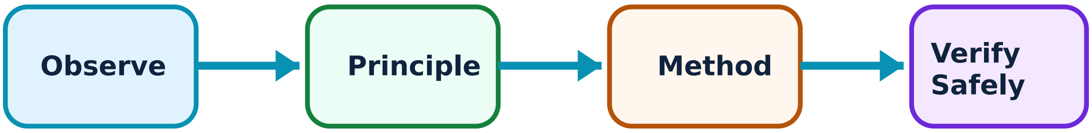
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Bioenergy, Storage and Hydrogen Economy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bioenergy and biofuels−

Bioenergy is derived from organic matter (biomass). Production technologies include anaerobic digestion for biogas, transesterification for biodiesel and fermentation for bioethanol, providing renewable alternatives for heating and transport.

Study and application method

Sketch a simple biogas digester and label the stages of anaerobic digestion; list three feedstocks suitable for biodiesel production.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a simple biogas digester and label the stages of anaerobic digestion; list three feedstocks suitable for biodiesel production.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Biofuel production must not compete with food security or lead to deforestation. Use only sustainable, non-food feedstocks for large-scale bioenergy.

2. Energy storage and hydrogen−

Energy storage (batteries, flywheels, thermal) is critical for integrating intermittent renewables into the grid. Hydrogen acts as a clean energy carrier, produced through electrolysis and used in fuel cells to generate electricity with only water as a by-product.

Study and application method

Compare the characteristics of a Li-ion battery and a supercapacitor; explain the concept of a 'green hydrogen' economy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the characteristics of a Li-ion battery and a supercapacitor; explain the concept of a 'green hydrogen' economy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Hydrogen is highly flammable and requires specialized storage and handling. All hydrogen systems must meet authorised safety standards for pressure and leakage.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Energy Economics, Sustainability and LCA

Energy demand characteristics; energy storage parameters; energy economics and policy; sustainability principles in energy; Life Cycle Assessment (LCA) of energy systems; carbon capture and storage (CCS).

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

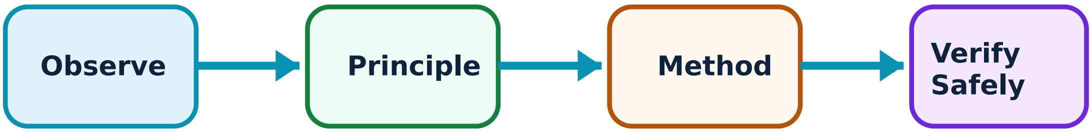
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Economics, Sustainability and LCA: identify the system, state its principle, follow the correct method and verify the result safely.

1. Energy economics and sustainability−

Energy economics analyzes the cost-effectiveness of different energy sources, including Levelized Cost of Energy (LCOE). Sustainability involves meeting current energy needs without compromising future generations, considering social, economic and environmental pillars.

Study and application method

Define 'Levelized Cost of Energy' (LCOE) and list four indicators of a sustainable energy system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Levelized Cost of Energy' (LCOE) and list four indicators of a sustainable energy system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Economic models assume certain discount rates and fuel prices. Do not rely on generic economic studies for project-specific investment decisions.

2. Life Cycle Assessment (LCA) and CCS−

LCA evaluates the environmental impact of an energy system from 'cradle to grave' (raw material extraction to disposal). Carbon Capture and Storage (CCS) technologies aim to capture CO2 from industrial sources and store it underground to reduce emissions.

Study and application method

Outline the major stages in a Life Cycle Assessment for a solar PV panel; describe the purpose of Carbon Capture and Storage (CCS) in a transition economy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Outline the major stages in a Life Cycle Assessment for a solar PV panel; describe the purpose of Carbon Capture and Storage (CCS) in a transition economy.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: LCA results are sensitive to system boundaries and data quality. CCS technologies are complex and involve long-term monitoring risks for leakage.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Energy Scenario and Solar Technologies

Use the concept flow to revise observation, principle, method and verification.

Module 2: Wind, Hydro and Geothermal Energy

Use the concept flow to revise observation, principle, method and verification.

Module 3: Bioenergy, Storage and Hydrogen Economy

Use the concept flow to revise observation, principle, method and verification.

Module 4: Energy Economics, Sustainability and LCA

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare three renewable energy sources in a conceptual table showing advantages and limitations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the components of a grid-connected solar PV system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the potential wind power conceptually given wind speed and turbine diameter.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a flow chart for the production of biodiesel from vegetable oil.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the stages of a Life Cycle Assessment (LCA) for a wind turbine.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Energy Scenario and Solar Technologies

Explain Energy trends and fossil fuel impacts and state one correct method, application or precaution.−

The world is shifting from carbon-intensive fossil fuels to renewable sources to mitigate climate change. Understanding macro trends in energy use (World and India) and the environmental consequences of fossil fuels is fundamental to the energy transition.

Answer framework: List three environmental impacts of fossil fuel use and compare the current renewable energy contribution in India vs the global average.

Important point: Energy data and climate projections are subject to frequent updates. Use current, authoritative reports (e.g., IEA, MNRE) for reliable energy planning.

Explain Solar thermal and PV systems and state one correct method, application or precaution.−

Solar energy is harnessed through thermal systems (water heaters, CSP) and Photovoltaic (PV) cells. PV technology converts sunlight directly into electricity using semiconductors. Efficiency depends on cell material, orientation and environmental factors.

Answer framework: Sketch a simple solar PV system and label the components; explain the difference between a solar thermal collector and a PV module.

Important point: Solar system performance is highly site-dependent. Do not design systems without authorised solar-path analysis and local weather data.

Module 2 — Wind, Hydro and Geothermal Energy

Explain Wind and hydro power and state one correct method, application or precaution.−

Wind energy uses turbines to convert kinetic energy into electricity, requiring sites with consistent wind speeds. Hydropower uses water flow to drive turbines, with pumped storage acting as a large-scale energy battery for grid stability.

Answer framework: Compare the working principles of a horizontal-axis wind turbine and a Pelton wheel hydroturbine; list the factors for selecting a wind-farm site.

Important point: Wind and hydro projects have significant land-use and ecological impacts. All projects must undergo authorised environmental and social impact assessments.

Explain Geothermal energy systems and state one correct method, application or

precaution.−

Geothermal energy harnesses heat from the Earth's interior for power generation or direct heating. Systems include dry steam, flash steam and binary cycle plants, depending on the temperature and state of the geothermal resource.

Answer framework: Describe the three main types of geothermal power plants and identify the regions in India with potential for geothermal energy.

Important point: Geothermal extraction can trigger seismic activity or release underground gases. Exploration and production must follow strict safety and environmental protocols.

Module 3 — Bioenergy, Storage and Hydrogen Economy

Explain Bioenergy and biofuels and state one correct method, application or precaution.−

Bioenergy is derived from organic matter (biomass). Production technologies include anaerobic digestion for biogas, transesterification for biodiesel and fermentation for bioethanol, providing renewable alternatives for heating and transport.

Answer framework: Sketch a simple biogas digester and label the stages of anaerobic digestion; list three feedstocks suitable for biodiesel production.

Important point: Biofuel production must not compete with food security or lead to deforestation. Use only sustainable, non-food feedstocks for large-scale bioenergy.

Explain Energy storage and hydrogen and state one correct method, application or precaution.−

Energy storage (batteries, flywheels, thermal) is critical for integrating intermittent renewables into the grid. Hydrogen acts as a clean energy carrier, produced through electrolysis and used in fuel cells to generate electricity with only water as a by-product.

Answer framework: Compare the characteristics of a Li-ion battery and a supercapacitor; explain the concept of a 'green hydrogen' economy.

Important point: Hydrogen is highly flammable and requires specialized storage and handling. All hydrogen systems must meet authorised safety standards for pressure and leakage.

Module 4 — Energy Economics, Sustainability and LCA

Explain Energy economics and sustainability and state one correct method, application or precaution.−

Energy economics analyzes the cost-effectiveness of different energy sources, including Levelized Cost of Energy (LCOE). Sustainability involves meeting current energy needs without compromising future generations, considering social, economic and environmental pillars.

Answer framework: Define 'Levelized Cost of Energy' (LCOE) and list four indicators of a sustainable energy system.

Important point: Economic models assume certain discount rates and fuel prices. Do not rely on generic economic studies for project-specific investment decisions.

Explain Life Cycle Assessment (LCA) and CCS and state one correct method, application or precaution.−

LCA evaluates the environmental impact of an energy system from 'cradle to grave' (raw material extraction to disposal). Carbon Capture and Storage (CCS) technologies aim to capture CO2 from industrial sources and store it underground to reduce emissions.

Answer framework: Outline the major stages in a Life Cycle Assessment for a solar PV panel; describe the purpose of Carbon Capture and Storage (CCS) in a transition economy.

Important point: LCA results are sensitive to system boundaries and data quality. CCS technologies are complex and involve long-term monitoring risks for leakage.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Energy Scenario and Solar Technologies.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Wind, Hydro and Geothermal Energy.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Bioenergy, Storage and Hydrogen Economy.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Energy Economics, Sustainability and LCA.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Global and Indian energy scenarios; impact of fossil fuels and climate change; introduction to renewable energy; solar radiation; solar thermal systems; solar photovoltaic (PV) technology and efficiency..
2. Develop a complete answer or practical plan for: Wind energy fundamentals; wind turbine design and performance; hydropower principles; hydroturbine selection; pumped hydro storage; geothermal energy sources and power systems..
3. Develop a complete answer or practical plan for: Bioenergy and biofuels (feedstocks, production technologies); biogas and biodiesel; energy storage technologies (mechanical, electrochemical, thermal); fuel cells and the hydrogen energy economy..
4. Develop a complete answer or practical plan for: Energy demand characteristics; energy storage parameters; energy economics and policy; sustainability principles in energy; Life Cycle Assessment (LCA) of energy systems; carbon capture and storage (CCS)..

LAST-MILE REVISION

Quick Revision

Module 1: Energy Scenario and Solar Technologies

Global and Indian energy scenarios; impact of fossil fuels and climate change; introduction to renewable energy; solar radiation; solar thermal systems; solar photovoltaic (PV) technology and efficiency.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Wind, Hydro and Geothermal Energy

Wind energy fundamentals; wind turbine design and performance; hydropower principles; hydroturbine selection; pumped hydro storage; geothermal energy sources and power systems.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Bioenergy, Storage and Hydrogen Economy

Bioenergy and biofuels (feedstocks, production technologies); biogas and biodiesel; energy storage technologies (mechanical, electrochemical, thermal); fuel cells and the hydrogen energy economy.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Energy Economics, Sustainability and LCA

Energy demand characteristics; energy storage parameters; energy economics and policy; sustainability principles in energy; Life Cycle Assessment (LCA) of energy systems; carbon capture and storage (CCS).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Energy trends and fossil fuel impacts	The world is shifting from carbon-intensive fossil fuels to renewable sources to mitigate climate change.
Solar thermal and PV systems	Solar energy is harnessed through thermal systems (water heaters, CSP) and Photovoltaic (PV) cells.
Wind and hydro power	Wind energy uses turbines to convert kinetic energy into electricity, requiring sites with consistent wind speeds.
Geothermal energy systems	Geothermal energy harnesses heat from the Earth's interior for power generation or direct heating.
Bioenergy and biofuels	Bioenergy is derived from organic matter (biomass).
Energy storage and hydrogen	Energy storage (batteries, flywheels, thermal) is critical for integrating intermittent renewables into the grid.
Energy economics and sustainability	Energy economics analyzes the cost-effectiveness of different energy sources, including Levelized Cost of Energy (LCOE).
Life Cycle Assessment (LCA) and CCS	LCA evaluates the environmental impact of an energy system from 'cradle to grave' (raw material extraction to disposal).

LEARNING RESOURCES

References

Recommended renewable-energy references

1. S. P. Sukhatme and J. K. Nayak, Solar Energy.
2. G. D. Rai, Non-Conventional Energy Sources.
3. B. H. Khan, Non-Conventional Energy Resources.
4. Twidell and Weir, Renewable Energy Resources.

Approved public renewable-energy sources

- https://nptel.ac.in/courses/112106318
- https://onlinecourses.nptel.ac.in/noc25_hs86/preview

These sources provide the verified study basis while official Course 6014A detail is unavailable; they do not replace official energy audits, authorised power-plant designs, official environmental reports or legal energy mandates.